Exercise

Uncle wants to write a recursive function to help him count how many ways he can sell n masks. It takes n as input and returns the number of ways to sell n masks. Help him complete the function by filling in the body of the function (you do not have to write the function header again in your answer). A non-recursive function will receive 0 marks. Since Uncle Tan does not sell many masks each day, you do not have to worry about (I) stack overflow, (ii) integer overflow, and (iii) the efficiency of your program.

In a Game, every participant chooses 6 numbers from 1 to 49. In each draw, 6 winning numbers and 1 additional number are drawn. The prize category for a participant is determined by how many of the chosen numbers match with the winning numbers and the additional number as shown in the table below.

(For simplicity's sake, we only consider the top-3 prize categories.)

Category	Numbers matched
1	6 winning numbers
2	5 winning numbers + additional number
3	5 winning numbers

You may assume that 1) the 6 chosen numbers and 7 drawn numbers have been read in and stored in two int arrays called chosen and drawn, respectively, and 2) the last number in drawn is the additional number.

Sample runs:

```
$ Enter your 6 chosen numbers: 15 3 29 2 42 40
$ Enter 7 drawn numbers: 42 2 15 40 3 29 30
Prize 1!
```

Matched winning numbers: 15, 3, 29, 2, 42, 40.

```
$ Enter your 6 chosen numbers: 15 3 29 2 42 40
$ Enter 7 drawn numbers: 3 15 25 40 2 42 29
Prize 2!
```

Matched winning numbers: 15, 3, 2, 42, 40. Matched additional number: 29.

```
$ Enter your 6 chosen numbers: 15 3 29 2 42 40
$ Enter 7 drawn numbers: 25 3 2 29 15 40 30
Prize 3!
```

Matched winning numbers: 15, 3, 29, 2, 40. Matched additional number: Nil.

```
$ Enter your 6 chosen numbers: 15 3 29 2 42 40
$ Enter 7 drawn numbers: 15 42 2 29 14 7 31
No prize.
```

Matched winning numbers: 15, 29, 2, 42. Matched additional number: Nil.

You are required to write a C program that stores and processes passenger data for a city's bus service using a **dynamically allocated 2D array** named bus[][]. The rows of the array represent **bus routes**, and the columns represent **days of the week** on which the buses operate. Each element of the array stores the **number of**

passengers who travelled on a specific route on a specific day. Your program must use **DMA (malloc)** to create the 2D array and must use **functions** for input, output, and all calculations.

	Mon	Tue	Wed	Thu	Fri
Route 0	8	12	9	7	10
Route 1	5	7	3	0	4
Route 2	20	15	18	21	14
Route 3	6	9	5	8	11

a. Write a function to input the passenger data into the dynamically allocated 2D array.
The data must be entered column-wise (day by day).
b. Write a function to display the passenger table exactly in matrix form.
c. Write a function to calculate and print the total number of passengers for Monday.
d. Write a function to calculate and print the total passengers for the route stored in row 0.
e. Write a function to find and print the maximum number of passengers for the route in row 3.
f. Write a function to find and print the minimum number of passengers for Thursday.
g. Write a function to calculate and print the average number of passengers for all routes and all days.
h. Write a function to find the route (row index) that has the highest total number of passengers, and print both:
1. the maximum total passengers, and
2. the row index in which it occurs.

Question No. 2

You are hired as a software Engineer in young's generation software house. You are hired to design an Ecommerce website. The functionality of E-commerce websites is as follows:
Customer can select multiple items from the list. The list is as shown below:

Electronic Items	**Beauty Items**	**Home Accessories**
Mobile (45K)	Foundation (7K)	Wall Paper (20K)
Television (100K)	Blush (5K)	Vase (5K)
Refrigerator (100K)	Lipstick (2K)	Flower (2K)
Microwave (100K)	Concealer (2K)	Decoration Pieces (1K)

The customer can select 1 or more than 1 item from the given list. In the bracket the price of each item is given.
Next, customer will put the selected item in the cart (you will ask the customer before putting the items in the cart).
You will display the items purchased by the client, the total of each item and the accumulated total.
Requirement.
You are required to make a structure array which will accept 5 customers' data. The structure array will have the customer ID and the category (Bold are the categories) mentioned in the above table.
You are required to make function for each process ie., input(), additem(),totalCost(), display().